Q① Count Digits in a number

Input n = 54321
Output - 5

→

```
Class Demo
{
  Public static void main(String args[])
  {
    int n = 54321;
    int count = 0;

    while (n > 0)
    {
      count++;
      n = n/10;
    }

    System.out.println(count);
  }
}
```

Q.② Extract all Digits.

input . num = 5382

→

```
class Demo
{
  Public static void main(String args[])
  {
    int num = 5382;
```

```
while (num>0)
{
    int rem = num%10;
    System.out.println(rem);
    num = num/10;
}
}
}
```

Q.③ Reverse a number

Input - 12345
Output - 54321

```
class Demo
{
  public static void main(String args[])
  {
    int num = 12345;
    int rev = 0;

    while (num>0)
    {
      int rem = num%10;
      rev = (rev*10) + rem;
      num = num/10;
    }
    System.out.println(rev);
  }
}
```

Q.4 Palindrome Number

Input - 121
Output - Palindrome

```
class Demo
{
  public static void main (String args[])
  {
    int num= 121;
    int temp= num;
    int rev = 0;
    while (temp >0)
    {
      ~~int temp=~~
      int rem= temp % 10;
      rev= (rev* 10) + rem;
      temp= temp/10;
    }
    if (num== rev)
    {
      SOP("~~No~~ Palindrome");
    }else{
      SOP(" Not Palindrome");
    }
  }
}
```

Q.① Sum of digits -

Input - 543
Output - 12

```
Class Demo
{
    public static void main(String args[])
    {
        int num = 543;
        int sum = 0;

        while (num > 0)
        {
            int rem = num % 10;
            sum = sum + rem;
            num = num / 10;
        }

        System.out.println(sum);
    }
}
```

Q.(8) Armstrong Number

Input - 153;
Output - Armstrong number.

→

```
Class Demo
{
  public static void main (String args[])
  {
      int num = 153;
      int temp = num;
      int count = 0;
      while (temp > 0)
      {
          count ++;
          temp = temp / 10;
      }
      int sum = 0;
      int temp2 = num;
      while ( temp2 > 0)
      {
          int rem = temp2 % 10;
          int rem2 = 1;
          for (int i = 0 ; i < count; i++ )
          {
              rem2 = rem2 * rem;
          }
          sum = sum + rem2;
      }
```

```
if (sum == num)
{
    SOP("Armstrong NO: ");
} else {
    SOP("Not Armstrong NO: ");
}
}
}
```

Q(1) Print All Divisors.

```
Class Demo
{
  public Static void main (String args[])
  {
    int num = 12;
    for (int i=1 ; i<= Math.sqrt(num); i++)
    {
      if (num%i = 0)
      {
        SOP (i);
        SOP (num/i);
      }
    }
  }
}
```

Method II

```
Class Demo
{
    Public Static void main (String args[])
    {
        int num =12;
        for(int i=1; i<=num; i++)
        {
            if (num%i==0)
            {
                SOP (i);
            }
        }
    }
}
```

Q.8 Count Number of Divisor -

```
Class Demo
{
    public static void main(String args[])
    {
        int num=12;
        int count=0;
        for(int i=1; i<=num; i++)
        {
            if (num%i==0)
            {
                count++;
            }
        SOP (count);
    }
```

Q① check prime Number

```
class Demo
{
    public static void main (String args[])
    {
        int num = 7;
        int count = 0;

        for(int i=1; i<=num ; i++)
        {
            if (num%i==0)
            {
                count ++;
            }
        }
        if (count==2)
        {
            SOP ("Prime");
        }else{
            SOP ("Not prime");
        }
    }
}
```

Q.⑩ Print All Prime Numbers up to N

```
class Demo
{
  Public static void main(String args[])
  {
    int num=20;

    for(int i=1; i<=num ; i++)
    { int count = 0;
      for (int j=1; j≤i ; j++)
      {
        if(i%j==0)
        {
          count++;
        }
      }
      ~~if(count==2)~~ if (count==2)
      ~~{~~
        ~~SOP(i);~~ { SOP(i);
      ~~}~~ }

    }
  }
}
```

Q. GCD of two numbers

```
class Demo
{
  public static void main (String args[])
  {
    int a=12;
    int b=18;
    int max=1;

    for (int i=1; i< Math.min(a,b); i++)
    {
      if (a%i==0 && b%i==0)
      {
        max=i;
      }
      Sop(max);
    }
  }
```

Q Check coprime or not

```
class Demo
{
  public static void main (String args[])
  {
    int a=8;
    int b=15;
    int max=1;
```

```
for (int i=2; i<Math.min (a,b); i++)
 {
    if (a%i==0 && b%i==0)
    {
        max=i;
    }
 }
 if (max==1)
 {
    SOP ("coprime");
 } else{

    SOP ("Not coprime");
    }
 }

}
```

Q. Check perfect Number

Input = 6.

```
class Demo
{
  public static void main (String args[])
  {
    int num=6;
    int sum=0;
    for(int i=1 ; i<num/2 ; i++)
    {
      if(num%i==0)
      {
        sum= sum+i;
      }
    }
    if(num==sum)
    {
      SOP("perfect No:")
    }else{
      SOP ("Not perfect No:")
    }
  }
}
```

Q.

```
class Demo
{
    Public static void main (String args[]
    {
        int a = 48;
        int b = 18;

        while (b!=0)
        {
            int temp = b;
            b = a%b;
            a = temp;
        }
        System.out.println(a);
    }
}
```